

Rational levels

Model — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Package

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Archimedean.** For all $p, q, r \in \Delta(X)$: if $p \succ q \succ r$ then there are $\alpha, \beta \in (0,1)$ with $\alpha p + (1-\alpha)r \succ q$ and $q \succ \beta p + (1-\beta)r$.
- **¬ Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \succcurlyeq r\}$ and $\{\lambda \in [0,1] : r \succcurlyeq \lambda p + (1-\lambda)q\}$ are closed.

0.2 Construction

Let $X = \{a, b, c\}$ and define a three-valued function V on $\Delta(X)$: $V(p) = 2$ if $p(a) \in \mathbb{Q}$; $V(p) = 1$ if $p(a) \notin \mathbb{Q}$ and $p(b) \in \mathbb{Q}$; $V(p) = 0$ otherwise. Let $p \succcurlyeq q$ iff $V(p) \geq V(q)$. This is a weak order.

Mixture continuity fails: along the line from δ_a to δ_b the mixture $\lambda\delta_a + (1-\lambda)\delta_b$ has $V = 2$ exactly when $\lambda \in \mathbb{Q}$, so $\{\lambda : \lambda\delta_a + (1-\lambda)\delta_b \succcurlyeq \delta_c\} = \mathbb{Q} \cap [0,1]$ is not closed.

Archimedean holds: a chain $p \succ q \succ r$ forces $V(p) = 2$, $V(q) = 1$, $V(r) = 0$, so $p(a) \in \mathbb{Q}$ and $r(a) \notin \mathbb{Q}$, in particular $p(a) \neq r(a)$. Along $m(\alpha) = \alpha p + (1-\alpha)r$ the coordinate $m(\alpha)(a) = r(a) + \alpha(p(a) - r(a))$ is rational for a dense set of α , so some $\alpha \in (0,1)$ close to 1 has $V(m(\alpha)) = 2 > V(q)$. For the other half, $m(\beta)(a)$ is irrational for all but countably many β ; and $m(\beta)(b) = r(b) + \beta(p(b) - r(b))$ is irrational for all but countably many β when $p(b) \neq r(b)$, and equals $r(b) \notin \mathbb{Q}$ when $p(b) = r(b)$ (because $V(r) = 0$). So some $\beta \in (0,1)$ close to 0 has $V(m(\beta)) = 0 < V(q)$.

0.3 Notes

The construction is deliberately artificial: it shows that the Archimedean axiom cannot replace mixture continuity even for weak orders, unless something like independence is added. Not yet checked by a human.

Record: topics/decision-theory/models/rational-levels.yaml